

Ruqi Zhang

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Education

Cornell University <i>Ph.D. in Statistics</i> Committee: Christopher De Sa, Thorsten Joachims, Giles Hooker	Ithaca, NY 2016–2021
Cornell University <i>M.S. in Computer Science</i>	Ithaca, NY 2019–2020
Renmin University of China <i>B.S. in Mathematics</i>	Beijing, China 2012–2016

Professional Experience

Purdue University <i>Assistant Professor, Computer Science</i>	West Lafayette, IN 2022–Present
University of Texas at Austin <i>Postdoctoral Fellow, Institute for Foundations of Machine Learning</i>	Austin, TX 2021–2022
Microsoft Research <i>Research Intern</i>	New England 2020
Microsoft Research <i>Research Intern</i>	Cambridge, UK 2019

Publications

G: Graduate student (primary advisor); U: Undergraduate student (primary advisor).

Conference Publications

- [C1] C. Cui, J. Zhang, Y. Zhou, J. Peng, S.-Y. Park, Z. Yang, P. Sankaranarayanan, **R. Zhang**, and Z. Wang. “Vilad: A large vision language diffusion framework for end-to-end autonomous driving”. In: *International Conference on Intelligent Robots and Systems* (2026).
- [C2] Y. Ding^G, Z. Qiu, B. Li^G, and **R. Zhang**. “Learning Self-Correction in Vision-Language Models via Rollout Augmentation”. In: *International Conference on Machine Learning* (2026).
- [C3] P. Jutras-Dube^G, J. Zhang, Z. Wang, and **R. Zhang**. “One-Step Diffusion Samplers via Self-Distillation and Deterministic Flow”. In: *International Conference on Artificial Intelligence and Statistics* (2026).
- [C4] J. Kim, W. Chen, D. Soleymanzadeh, Y. Ding^G, X. Gao, Z. Tu, **R. Zhang**, F. Fei, S. Veer, Y. Lyu, et al. “Modular Safety Guardrails Are Necessary for Foundation-Model-Enabled Robots in the Real World”. In: *International Conference on Machine Learning* (2026).
- [C5] H. Lin^U and **R. Zhang**. “Gradient-Guided Reward Optimization for Inference-time Alignment”. In: *Proceedings of the Conference on Uncertainty in Artificial Intelligence (UAI)*. 2026.
- [C6] P. Mohanty and **R. Zhang**. “Slithering Through Gaps: Capturing Discrete Isolated Modes via Logistic Bridging”. In: *International Conference on Artificial Intelligence and Statistics* (2026).
- [C7] P. Pynadath^G, J. Shi, and **R. Zhang**. “CANDI: Hybrid Discrete-Continuous Diffusion Models”. In: *International Conference on Machine Learning* (2026).
- [C8] Z. Wang, S. R. Kasa, S. K. Kasa, J. Zou, S. Negi, **R. Zhang**, N. Jiang, Q. Song, et al. “DIVERSED: Relaxed speculative decoding via dynamic ensemble verification”. In: *International Conference on Artificial Intelligence and Statistics* (2026).

- [C9] H. Zheng, **R. Zhang**, and G. Lin. “Exploring Non-Convex Discrete Energy Landscapes: A Langevin-Like Sampler with Replica Exchange”. In: *The AAAI Conference on Artificial Intelligence* (2026).
- [C10] Y. Ding^G, B. Li^G, and **R. Zhang**. “ETA: Evaluating Then Aligning Safety of Vision Language Models at Inference Time”. In: *International Conference on Learning Representations* (2025).
- [C11] Y. Ding^G and **R. Zhang**. “Sherlock: Self-Correcting Reasoning in Vision-Language Models”. In: *Neural Information Processing Systems* (2025).
- [C12] B. Li^G, Y. Wang, A. Lochab^G, A. Grama, and **R. Zhang**. “Cascade reward sampling for efficient decoding-time alignment”. In: *Conference on Language Modeling* (2025).
- [C13] B. Li^G, Y. Wu, X. Luo, and **R. Zhang**. “Reward-Shifted Speculative Sampling Is An Efficient Test-Time Weak-to-Strong Aligner”. In: *The Conference on Empirical Methods in Natural Language Processing* (2025).
- [C14] Q. Liao^U, A. Lochab^G, and **R. Zhang**. “VERA-V: Variational Inference Framework for Jailbreaking Vision-Language Models”. In: *International Conference on Machine Learning* (2025).
- [C15] X. Liu^G, H. Du, W. Deng, and **R. Zhang**. “Optimal Stochastic Trace Estimation in Generative Modeling”. In: *International Conference on Artificial Intelligence and Statistics* (2025).
- [C16] A. Lochab^G, L. Yan, P. Pynadath^G, X. Zhang, and **R. Zhang**. “VERA: Variational Inference Framework for Jailbreaking Large Language Models”. In: *Neural Information Processing Systems* (2025).
- [C17] A. Lochab^G and **R. Zhang**. “Energy-Based Reward Models for Robust Language Model Alignment”. In: *Conference on Language Modeling* (2025).
- [C18] J. Lu, G. Rohit, A. Moradipari, K. Han, **R. Zhang**, and W. Ziran. “On Learning Closed-Loop Probabilistic Multi-Agent Simulator”. In: *International Conference on Intelligent Robots and Systems* (2025).
- [C19] X. Luo, C. S. Bai, B. Li^G, P. Drineas, **R. Zhang**, and B. Bullins. “Stacey: Promoting Stochastic Steepest Descent via Accelerated p-Smooth Nonconvex Optimization”. In: *International Conference on Machine Learning* (2025).
- [C20] V. Punyamoorthy, P. Jutras-Dube^G, **R. Zhang**, V. Aggarwal, D. Conover, and A. Bera. “Dynamic Obstacle Avoidance through Uncertainty-Based Adaptive Planning with Diffusion”. In: *International Conference on Intelligent Robots and Systems* (2025).
- [C21] P. Pynadath^G and **R. Zhang**. “Controlled LLM Decoding via Discrete Auto-regressive Biasing”. In: *International Conference on Learning Representations* (2025).
- [C22] Y. Wang, R. Chen, B. Li^G, D. Cho, Y. Deng, **R. Zhang**, T. Chen, Z. Wang, A. Grama, and J. Hong. “More is Less: The Pitfalls of Multi-Model Synthetic Preference Data in DPO Safety Alignment”. In: *Conference on Language Modeling* (2025).
- [C23] B. Zhang^G and **R. Zhang**. “CoT-UQ: Improving Response-wise Uncertainty Quantification in LLMs with Chain-of-Thought”. In: *Findings of The Annual Meeting of the Association for Computational Linguistics* (2025).
- [C24] P. Jutras-Dube^G, **R. Zhang**, and A. Bera. “Adaptive Planning with Generative Models under Uncertainty”. In: *International Conference on Intelligent Robots and Systems* (2024).
- [C25] B. Lei, D. Xu, **R. Zhang**, S. He, and B. K. Mallick. “Balance is Essence: Accelerating Sparse Training via Adaptive Gradient Correction”. In: *Conference on Parsimony and Learning* (2024).
- [C26] B. Lei, D. Xu, **R. Zhang**, and B. Mallick. “Embracing Unknown Step by Step: Towards Reliable Sparse Training in Real World”. In: *Transactions on Machine Learning Research* (2024).
- [C27] B. Li^G and **R. Zhang**. “Entropy-MCMC: Sampling from Flat Basins with Ease”. In: *International Conference on Learning Representations* (2024).
- [C28] J. Li^G, Z. Miao, Q. Qiu, and **R. Zhang**. “Training Bayesian Neural Networks with Sparse Subspace Variational Inference”. In: *International Conference on Learning Representations* (2024).

- [C29] X. Liu, B. Lei, **R. Zhang**, and D. Xu. “Adaptive Draft-Verification for Efficient Large Language Model Decoding”. In: *The AAAI Conference on Artificial Intelligence* (2024).
- [C30] T. Papamarkou, M. Skoularidou, **R. Zhang**, et al. “Position paper: Bayesian deep learning in the age of large-scale ai”. In: *International Conference on Machine Learning* (2024).
- [C31] P. Pynadath^G, R. Bhattacharya, A. Hariharan, and **R. Zhang**. “Gradient-based Discrete Sampling with Automatic Cyclical Scheduling”. In: *Neural Information Processing Systems* (2024).
- [C32] Z. Wang, Y. Chen, Q. Song, and **R. Zhang**. “Enhancing Low-Precision Sampling via Stochastic Gradient Hamiltonian Monte Carlo”. In: *Transactions on Machine Learning Research* (2024).
- [C33] K. Goshvadi, H. Sun, X. Liu, A. Nova, **R. Zhang**, W. S. Grathwohl, D. Schuurmans, and H. Dai. “DISCS: A Benchmark for Discrete Sampling”. In: *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*. 2023.
- [C34] T. Islam, **R. Zhang**, and D. Goldwasser. “Analysis of Climate Campaigns on Social Media using Bayesian Model Averaging”. In: *AAAI/ACM Conference on AI, Ethics, and Society* (2023). **Oral Presentation, top 11%**.
- [C35] B. Lei, **R. Zhang**, D. Xu, and B. Mallick. “Calibrating the Rigged Lottery: Making All Tickets Reliable”. In: *International Conference on Learning Representations* (2023).
- [C36] Y. Xiang^G, D. Zhu, B. Lei, D. Xu, and **R. Zhang**. “Efficient Informed Proposals for Discrete Distributions via Newton’s Series Approximation”. In: *International Conference on Artificial Intelligence and Statistics*. PMLR. 2023, pp. 7288–7310.
- [C37] W. Zhang and **R. Zhang**. “DP-Fast MH: Private, Fast, and Accurate Metropolis-Hastings for Large-Scale Bayesian Inference”. In: *International Conference on Machine Learning* (2023).
- [C38] D. Zhu, B. Lei, J. Zhang, Y. Fang, Y. Xie, **R. Zhang**, and D. Xu. “Rethinking data distillation: Do not overlook calibration”. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2023, pp. 4935–4945.
- [C39] **R. Zhang**, Q. Liu, and X. Tong. “Sampling in Constrained Domains with Orthogonal-Space Variational Gradient Descent”. In: *Advances in Neural Information Processing Systems* 35 (2022), pp. 37108–37120.
- [C40] **R. Zhang**, X. Liu, and Q. Liu. “A Langevin-like Sampler for Discrete Distributions”. In: *International Conference on Machine Learning*. 2022, pp. 26375–26396.
- [C41] **R. Zhang**, A. G. Wilson, and C. De Sa. “Low-Precision Stochastic Gradient Langevin Dynamics”. In: *International Conference on Machine Learning*. PMLR. 2022, pp. 26624–26644.
- [C42] **R. Zhang**, Y. Li, C. De Sa, S. Devlin, and C. Zhang. “Meta-Learning Divergences for Variational Inference”. In: *International Conference on Artificial Intelligence and Statistics*. PMLR. 2021, pp. 4024–4032.
- [C43] **R. Zhang**, A. F. Cooper, and C. De Sa. “AMAGOLD: Amortized Metropolis adjustment for efficient stochastic gradient MCMC”. In: *International Conference on Artificial Intelligence and Statistics*. PMLR. 2020, pp. 2142–2152.
- [C44] **R. Zhang**, A. F. Cooper, and C. M. De Sa. “Asymptotically optimal exact minibatch metropolis-hastings”. In: *Advances in Neural Information Processing Systems* 33 (2020). **Spotlight Presentation, top 3%**, pp. 19500–19510.
- [C45] **R. Zhang**, C. Li, J. Zhang, C. Chen, and A. G. Wilson. “Cyclical stochastic gradient MCMC for Bayesian deep learning”. In: *International Conference on Learning Representations*. **Oral Presentation, top 2%**. 2020.
- [C46] **R. Zhang** and C. M. De Sa. “Poisson-Minibatching for Gibbs Sampling with Convergence Rate Guarantees”. In: *Advances in Neural Information Processing Systems*. **Spotlight Presentation, top 2.5%**. 2019, pp. 4922–4931.

- [C47] **R. Zhang** and Z. Lu. “Large Scale Sparse Clustering.” In: *International Joint Conference on Artificial Intelligence*. 2016, pp. 2336–2342.

Journal Publications

- [J1] J. Zhang, J. Lu, X. Wu, Z. Wang, and **R. Zhang**. “Inference Acceleration of Autoregressive Normalizing Flows by Selective Jacobi Decoding”. In: *Transactions on Machine Learning Research* (2026). **Featured Certification, top 10%**.
- [J2] B. Li^G and **R. Zhang**. “Making Reliable and Flexible Decisions in Long-tailed Classification”. In: *Transactions on Machine Learning Research* (2025).
- [J3] M. Li^U and **R. Zhang**. “Reheated Gradient-based Discrete Sampling for Combinatorial Optimization”. In: *Transactions on Machine Learning Research* (2025).
- [J4] C. Zhuang^G, D. Mukherjee^G, Y. Lu, T. Fu, and **R. Zhang**. “Gradient GA: Gradient Genetic Algorithm for Drug Molecular Design”. In: *Transactions on Machine Learning Research* (2025).

Book Chapters

- [B1] W. Chen, B. Li^G, **R. Zhang**, and Y. Li. “Bayesian Computation in Deep Learning”. In: *In the Handbook of Markov Chain Monte Carlo* (2025).

Preprints

- [P1] P. Mohanty, R. Bhattacharya, and **R. Zhang**. “Entropy-Guided Sampling of Flat Modes in Discrete Spaces”. In: *arXiv preprint arXiv:2505.02296* (2025).

Awards and Honors

- AAAI New Faculty Highlights (2025)
- Amazon Research Award (2024)
- Seed for Success Acorn Award (2024)
- Ross-Lynn Research Scholar (2023)
- UAI Top Reviewer (2023)
- ICML Best Reviewer (Top 10%) (2021)
- Spotlight Rising Star in Data Science, University of Chicago (one of 10, 2020)
- NeurIPS Top 10% Reviewer Award (2020)
- NeurIPS Travel Grant (2019)

Talks

- Discovering and Controlling Safety Risks in Foundation Models: A Probabilistic Perspective, NSF NAIRR Regional AI Workshop, May 2026
- Gradient-Based Discrete Sampling: Algorithms and Applications, Flatiron Institute, May 2026
- Discovering and Controlling Safety Risks in Foundation Models: A Probabilistic Perspective, NCSU AI Seminar, April 2026
- Discovering and Controlling Safety Risks in Foundation Models: A Probabilistic Perspective, Purdue CERIAS Seminar, March 2026
- Discovering and Controlling Safety Risks in Foundation Models: A Probabilistic Perspective, Purdue ICON Seminar, February 2026
- The Future of Probabilistic Modeling in Data Mining and AI, IEEE International Conference on Data Mining Panel, November 2025
- Gradient-Based Discrete Sampling: Algorithms and Applications, Monte Carlo Seminar, October 2025
- Aligned and Safe LLMs via Probabilistic Modeling, IJCAI Workshop on User-Aligned Assessment of Adaptive AI Systems, August 2025

- Toward Capable and Reliable LLMs via Probabilistic Modeling, Midwest Machine Learning Symposium, June 2025
- Aligned and Reliable LLMs via Probabilistic Modeling, Morgan Stanley Machine Learning External Speaker Series, April 2025
- Transformative AI: LLMs, Robotics, and Scientific Innovation, Purdue CS Graduate Symposium Panel, March 2025
- Scalable and Efficient Probabilistic Inference for Bayesian Deep Learning and Generative Modeling, AAAI-25 New Faculty Highlights, March 2025
- New Advances in Discrete Sampling and Applications in LLMs, Purdue Statistics Faculty Retreat, August 2024
- Low-precision Sampling for Probabilistic Deep Learning, Invited talk at NeurIPS Workshop on Machine Learning with New Compute Paradigms, NeurIPS, December 2023
- Sampling in Discrete and Constrained Domains, Invited talk at ICML Workshop on Structured Probabilistic Inference & Generative Modeling, ICML, July 2023
- Scalable and Reliable Inference for Probabilistic Modeling, Invited talk at Center for Data Science and Machine Learning, National University of Singapore, October 2022
- A Langevin-like Sampler for Discrete Distributions, Spotlight presentation at ICML, July 2022
- Low-Precision Stochastic Gradient Langevin Dynamics, Spotlight presentation at ICML, July 2022
- Scalable and Reliable Inference for Probabilistic Modeling, Invited talk at Simons Institute, UC Berkeley, November 2021
- Asymptotically Optimal Exact Minibatch Metropolis-Hastings, Spotlight talk in Rising Stars in Data Science Workshop at University of Chicago, January 2021
- Asymptotically Optimal Exact Minibatch Metropolis-Hastings, Spotlight presentation at NeurIPS, December 2020
- Cyclical Stochastic Gradient MCMC for Bayesian Deep Learning, Oral presentation at ICLR, April 2020
- Poisson-Minibatching for Gibbs Sampling with Convergence Rate Guarantees, Spotlight presentation at NeurIPS, December 2019

Teaching

- **CS 37300** — Data Mining and Machine Learning, Purdue (Spring 2024, Spring 2026, Fall 2026)
- **CS 57800** — Statistical Machine Learning, Purdue (Spring 2023, Fall 2023, Fall 2024, Fall 2025)
- **CS 59200** — Probabilistic Machine Learning, Purdue (Fall 2022)
- **CS 57100** — Artificial Intelligence, Purdue (Guest Lecturer)
- Head TA, **CS 4820** — Algorithms, Cornell (Spring 2021)
- TA, **ILRST 5050** — Statistics at Work, Cornell (Fall 2018)
- TA, **STSCI 2100** — Introductory Statistics, Cornell (Spring 2018)
- TA, **MATH 3110** — Introduction to Analysis, Cornell (Spring 2017)
- TA, **STSCI 3110** — Probability Models and Inference for the Social Sciences, Cornell (Fall 2016, 2017, 2020)

Service

Chair

- Midwest Machine Learning Symposium (2026)

Organizer

- Symposium on Advances in Approximate Bayesian Inference (2022–2025)
- NeurIPS Workshop on ML with New Compute Paradigms (2024)
- ICML Workshop on Sampling and Optimization in Discrete Space (2023)
- ICML Women in Machine Learning (WiML) Un-Workshop on Safely Navigating Scalability-reliability Trade-offs in ML Methods (2021)

Area Chair

- ICML 2025-2026
- NeurIPS 2024–2026
- ICLR 2026
- AISTATS 2024–2026
- UAI 2024–2026

Editor

- Transactions on Machine Learning Research (2024-present)

Reviewer

- Journal of the American Statistical Association
- Statistics and Computing
- Entropy
- NeurIPS 2018–2023
- NeurIPS Workshop Proposal 2024–2025
- ICML 2019–2023 (expert reviewer in 2021)
- ICLR 2019–2025
- AISTATS 2020–2023
- AAAI 2020
- UAI 2019
- MLSys 2024

Panelist

- NSF Panels (2022–2026)
- DOE Panel (2024)

Mentor

- Women in Machine Learning (WiML), COLM 2025
- AISTATS Submission Mentoring Program 2022

Department

- Purdue CS PhD Admission Committee (2023–2026)

Grants

- **GPU-Accelerated Low-Precision Sampling for Scalable Statistical Inference**, NVIDIA Academic Grant Program, PI (100%), 30K H100 GPU-hour (2026)
- **Student Travel Grant for the 2026 Midwest Machine Learning Symposium**, NSF IIS RI, PI (100%), \$25K (2026)
- **REU Student Funding**, NSF CISE, PI (100%), \$10K (2026)
- **Uncertainty-Aware Decision-Making for Optimal Control Transitions in Shared Autonomy**, Toyota Research Institute, Co-PI (50%), \$783K (2026–2028)
- **AI-guided design of compact CRISPR-associated transposons for programmable gene insertion**, Purdue Biomolecular Design Seed Grant, Co-PI (25%), \$50K (2026–2027)
- **Multi-Modal Imaging and AI Framework for Advanced Visual Inspection of Pharmaceutical Vials**, Young Institute Pharmaceutical Manufacturing Consortium, Co-PI (10%), \$1400K (2025-2026)
- **Gradient-Based Discrete Markov Chain Monte Carlo: Algorithms, Theoretical Foundations, and Applications**, NSF IIS RI, PI (100%), \$600K (2025–2028)
- **Amazon Research Award**, Amazon, PI (100%), \$69,967 Cash + \$50,000 AWS (2025–2026)
- **Ross-Lynn Research Scholar Fund**, Purdue, PI (100%), 1 year GRA (2023–2024)

Student Advising

Postdoctoral Scholar

- Jiaru Zhang (co-advised with Ziran Wang), 2025–Present

Ph.D. Students (Primary Advisor)

- Bolian Li (CS), 2022–Present

- Pascal Jutras-Dube (CS), 2023–Present
- Patrick Pynadath (CS), 2023–Present
- Chris Zhuang (CS), 2024–Present
- Anamika Lochab (CS), 2024–Present
- Mohammad Areeb (CS), 2024–Present
- Yi Ding (CS), 2025–Present
- Anant Shyam (CS), 2025–Present

M.S. Students

- Debadyuti Mukherjee (CS), 2024–2025
- Kunal Kapur (CS), 2025

Undergraduate Students

- Jeremy Lu (CS), 2026–Present
- Ivan Philip (CS), 2024–2025
- Disha Sanjay Maheshwari (ECE), 2024–2025; now CS Ph.D. at Georgia Tech

Ph.D. Committee Member

- Yanran Wu (CS), 2023–Present
- Yifan Wang (CS), 2022–Present
- Yunxin Sun (CS), 2022–Present
- Shahab Rahimirad (CS), 2022–Present
- Chen Peng (CS), 2022–Present
- Xinyu Luo (CS), 2022–Present
- Shilong Lei (CS), 2022–Present
- Chenzhe Jin (CS), 2022–Present
- Ziyi Wang (Statistics), 2022–Present
- Zhaoqing Wu (CS), 2022–Present
- Han Zhu (CS), 2021–Present
- Yucheng Zhang (CS), 2021–Present
- Jinzhao Li (CS), 2021–Present
- Jonathan D. Rosenthal (CS), 2019–Present
- Tunazzina Islam (CS), 2019–2025
- Tinghan Yang (CS), 2018–Present

Interns

- Qilin Liao, Intern at Purdue (2025–2026); now CS Ph.D. student at Purdue
- Hankun Lin, Intern at Purdue (2025–2026); now CS Ph.D. student at Purdue
- Sujong Lee, Intern at Purdue (2025–2026); now CS Ph.D. student at Nanyang Technological University
- Yi Ding, Intern at Purdue (2024–2025); now CS Ph.D. student at Purdue
- Xinyang Liu, Intern at Purdue (2024–2025); now Stats Ph.D. student at UT Austin
- Boxuan Zhang, Intern at Purdue (2024–2025); now CS Ph.D. student at Rutgers University
- Junbo Li, Intern at Purdue (2023–2024); now CS Ph.D. student at UT Austin
- Muheng Li, Intern at Purdue (2023–2024); now Stats Ph.D. student at University of Toronto